

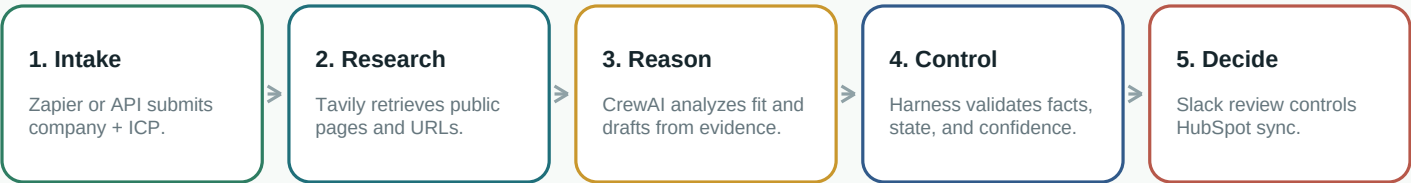
From account research to a reviewable decision

B2B teams repeatedly research target companies, judge whether they fit, write a first message, and copy the result into a CRM. This system turns that manual loop into a visible workflow without giving AI the final say.

The job to be done

For each target account, the workflow answers: **What is happening?** using public evidence; **Is this a fit?** using a configurable ideal customer profile; and **What should happen next?** using a draft that a person reviews. The result is not an auto-send sales bot. It is an AI research desk with a human approval counter.

One company through the system



Six choices that make the workflow dependable

CrewAI roles, deterministic control

Researcher, Analyst, and Writer handle uncertain reasoning. Python owns validation, status, retries, and database writes.

A worker for long tasks

FastAPI returns a tracking ID quickly. A database-backed worker claims the queued run and performs the slower agent work.

Evidence travels with claims

Every accepted company claim retains its source URL, type, retrieval time, and grounding result.

Human approval before CRM

Slack captures approve, reject, or revise. Only approved drafts become HubSpot Notes; no outreach is auto-sent.

One system of record

PostgreSQL stores runs, companies, findings, analysis, drafts, and the event timeline so the process remains inspectable.

Two modes for two jobs

Offline deterministic mode makes CI repeatable. CrewAI + Tavily mode demonstrates live agent and web-research behavior.

A complete workflow with an honest boundary

The value of the project is not that an LLM can write an email. The value is the surrounding operating system: evidence, state, review, observability, release controls, and an honest boundary between demonstrated behavior and enterprise scale.

Three layers

Business workflow

Zapier | Slack | HubSpot | Power BI

Lets non-technical users start work, make the decision, file approved output, and monitor the process.

Agent core

CrewAI | Tavily | ICP profiles

Collects public evidence, explains account fit, and prepares a draft with structured outputs.

Control harness

FastAPI | Worker | PostgreSQL | Validation

Owns lifecycle, evidence checks, retries, security boundaries, events, and long-running execution.

What was actually verified

82 tests

Offline CI suite

5 / 5

Grounded live findings

65

Verified fit score

15 runs

Power BI snapshot

The Oscar Health smoke test ran CrewAI + Tavily on Render, stored five grounded findings in Neon, reached Slack review, recorded a human approval, and created an associated HubSpot Note. GitHub protected pull requests with Ruff, unit tests, and a Docker build; deployment ran behind a test gate. Power BI refreshed from protected, schema-validated CSV exports.

Delivery model

AI-assisted implementation

AI coding agents produced the architecture and code. The owner configured Zapier, Render, Neon, Slack, and HubSpot; ran live workflows; reported failures; completed human approvals; and verified releases.

What enterprise scale would add

A durable queue, shared rate limits, centralized alerts and tracing, backup and restore drills, reviewer authorization, incident procedures, and scheduled dashboard refresh.

The core product rule remains unchanged at either scale: **AI prepares evidence and a draft; a person approves the business action.**